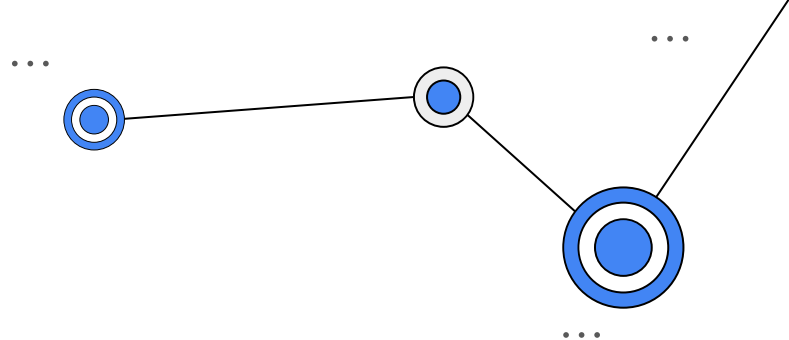




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440MI

DATA-DRIVEN SYSTEMS ENGINEERING

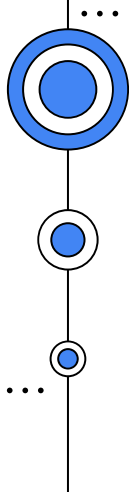
L17 - MLOps Operations

Sylvio Barbon Junior
sylvio.barbonjunior@units.it

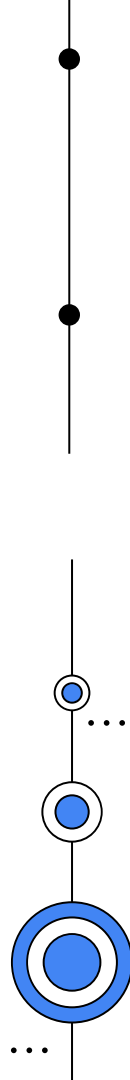
17

MLOps

MLOps Operations



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Last Lesson



AI Maturity
Themes



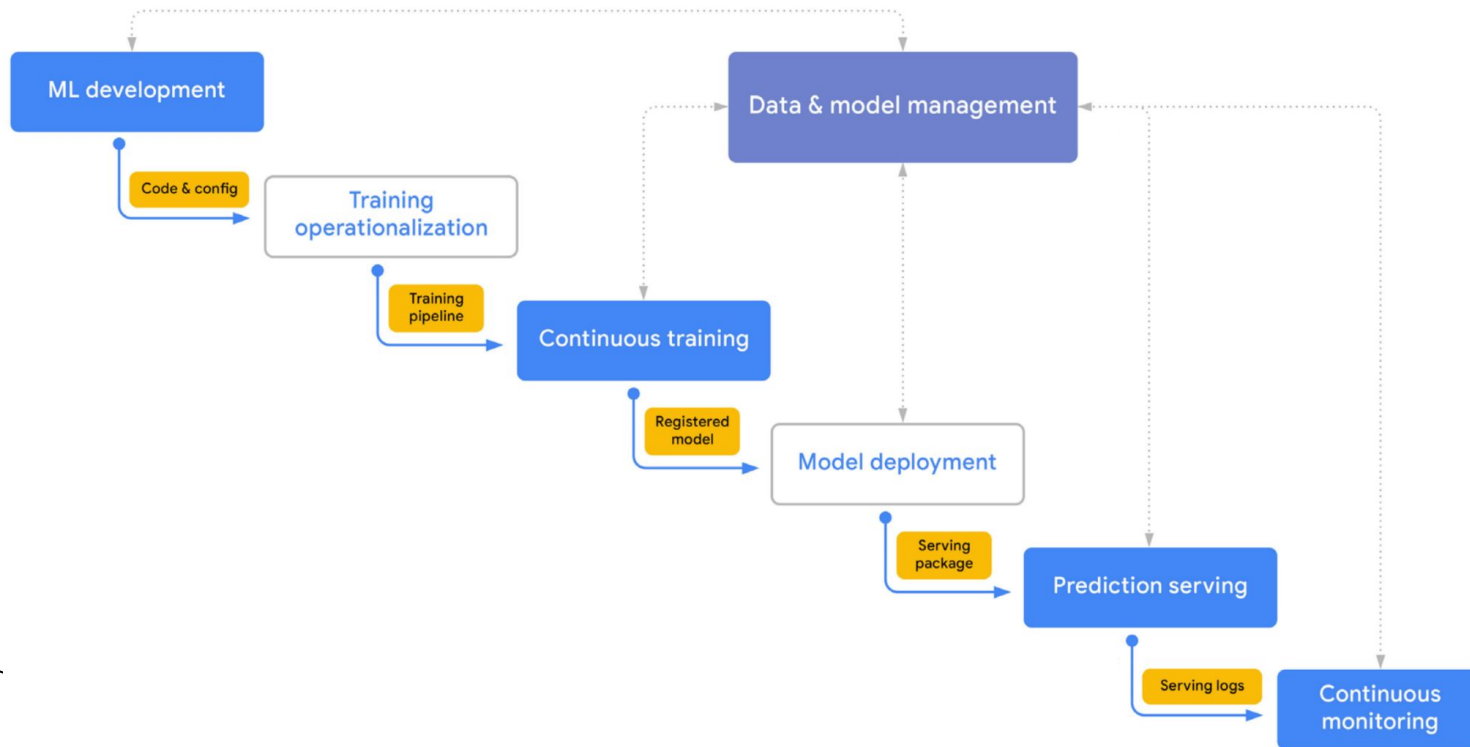
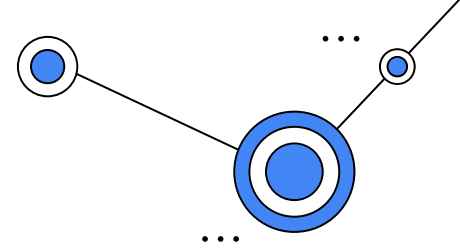
MLOps
Processes

Today's goal:

- MLOps Workflow
- MLOps Processes: Development
 - Questions;
 - Goals;
 - Good Practices;
- MPOps Project

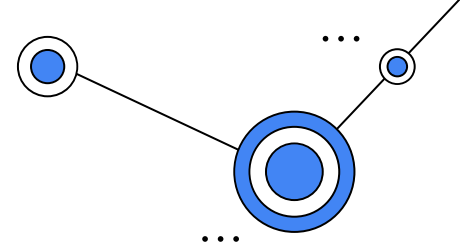
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MLOps Workflow



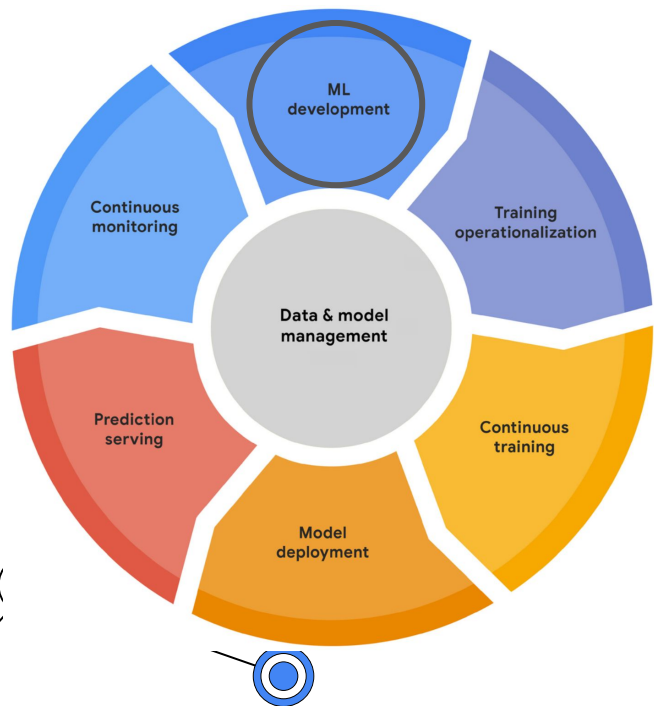
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MLOps Processes



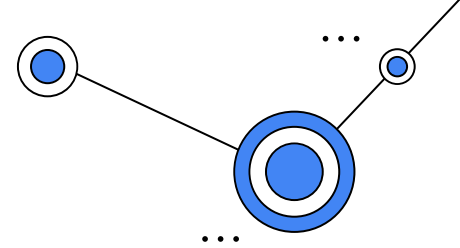
ML development

Experimentation is the core activity in ML development, where your data scientists can rapidly try several ideas for data preparation and ML modeling.



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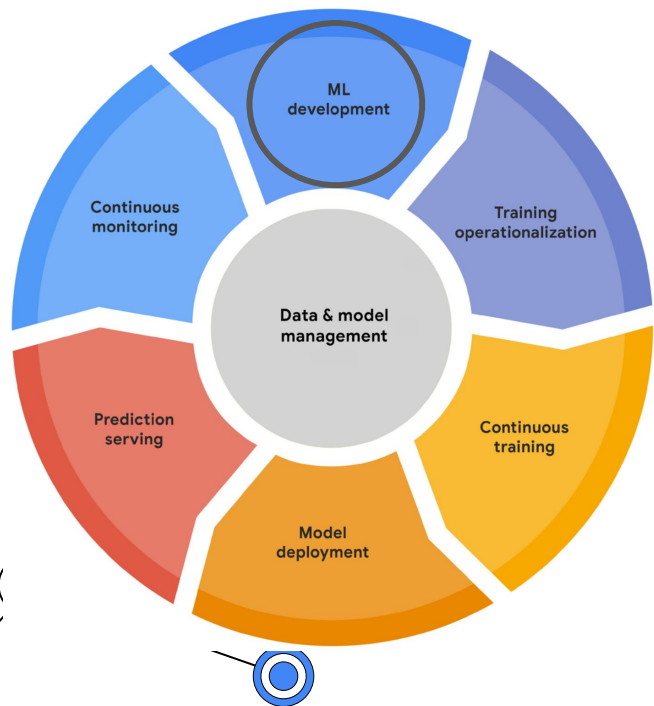
MLOps Processes



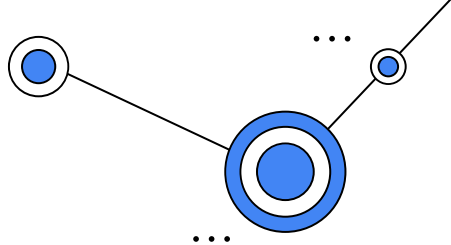
ML development

Experimentation starts when the ML use case is well defined, meaning that the following questions have been answered:

- **What is the task?**
- **How can we measure business impact?**
- **What is the evaluation metric?**
- **What is the relevant data?**
- **What are the training and serving requirements?**



MLOps Processes

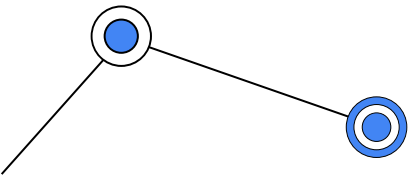


ML development

Experimentation aims to arrive at an **effective prototype model** for the ML use case at hand.

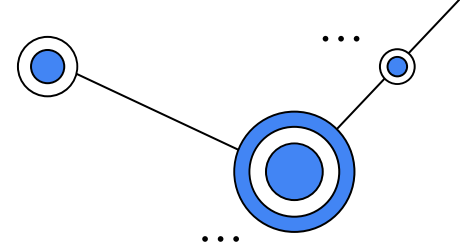
In addition to experimentation, **data scientists need to formalize** their ML training procedures.

They do this by implementing an **end-to-end pipeline**, so that the procedures can be operationalized and run in production.



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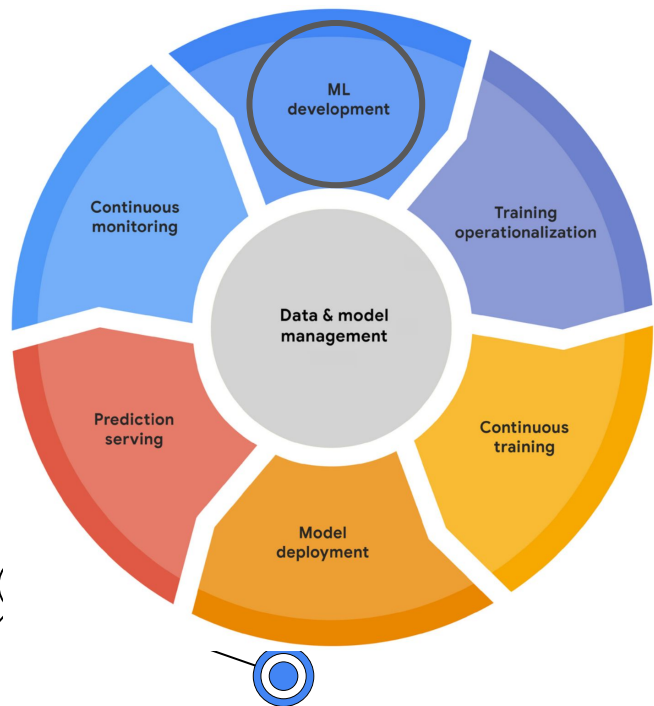
MLOps Processes



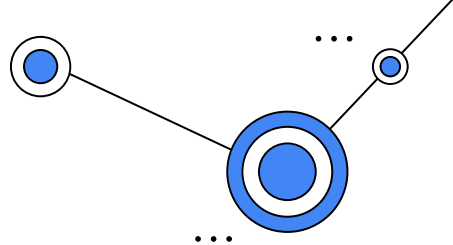
ML development

During **experimentation**, data scientists typically perform the following steps:

- Data discovery, selection, and exploration.
- Data preparation and feature engineering, using interactive data processing tools.
- Model prototyping and validation.



MLOps Processes

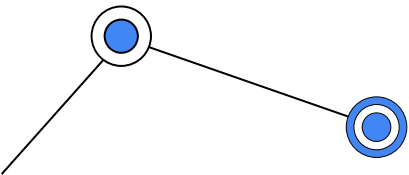


ML development

In general, the **key success aspects** for this process are:

- experiment tracking;
- reproducibility;
- collaboration;

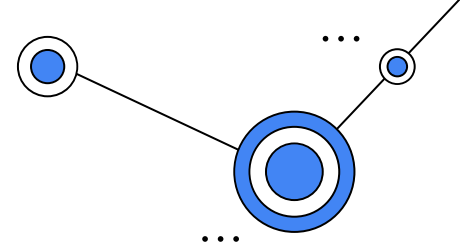
For example, when your data scientists begin working on an ML use case, it can save them time if they can find previous experiments that have similar use cases and that reproduce the results of those experiments;



Data scientists need to be able to compare various experiments and runs of the same experiment so that they understand the factors that lead to changing the model's predictive behavior and to performance improvements.

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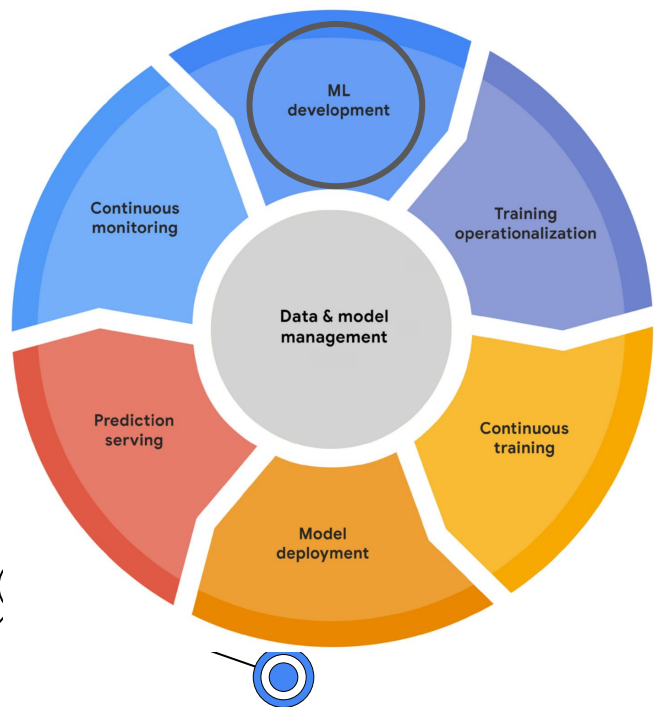
MLOps Processes



ML development

To be able to **reproduce** an **experiment**, including:

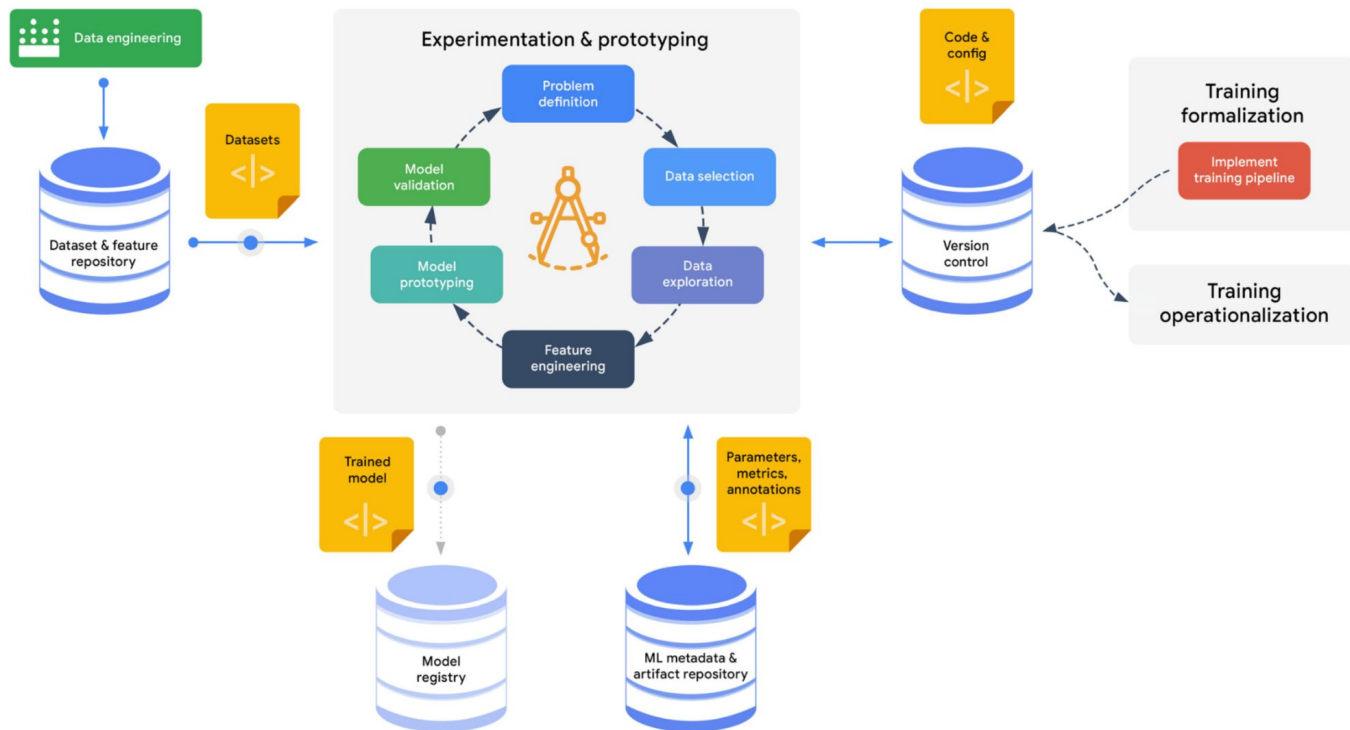
- A pointer to the version of the training code in the **version control system**.
- The **model architecture and pretrained modules** that were used.
- Hyperparameters, including trials of **automated hyperparameter tuning and model selection**.
- **Information about training, validation, and testing data splits** that were used.
- Model **evaluation metrics** and the validation procedure that was used.



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MLOps Processes

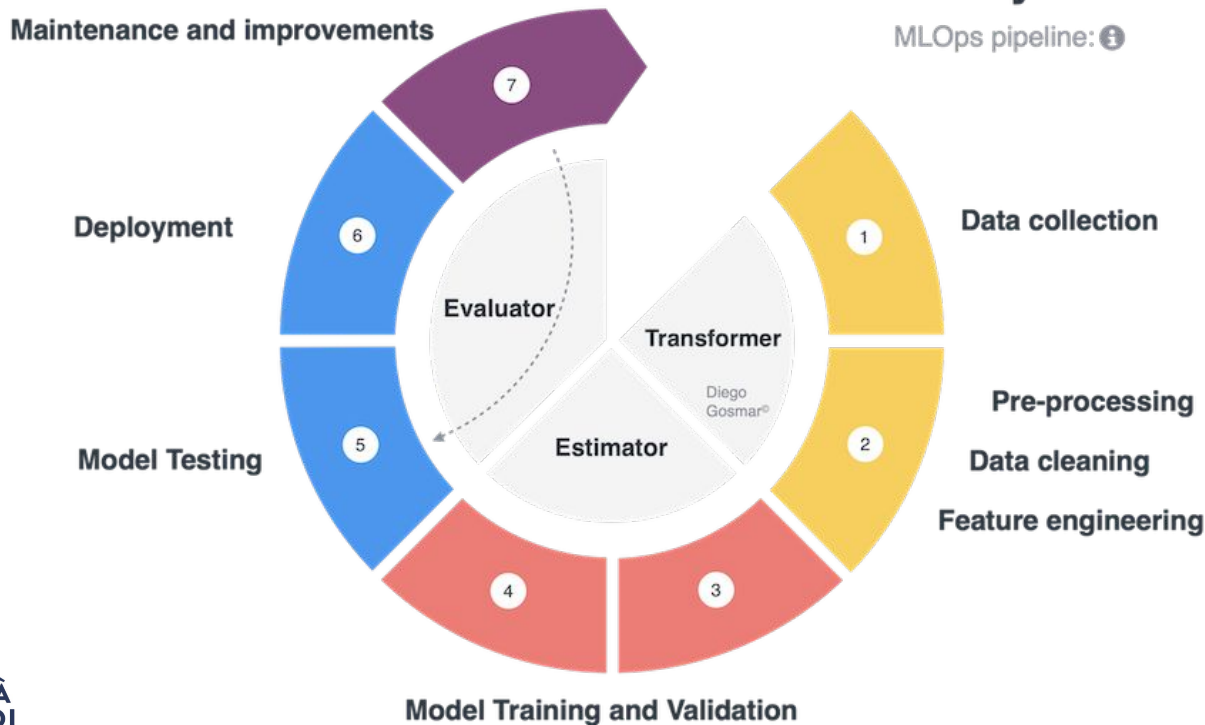
ML development



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Machine Learning life cycle

MLOps pipeline: 



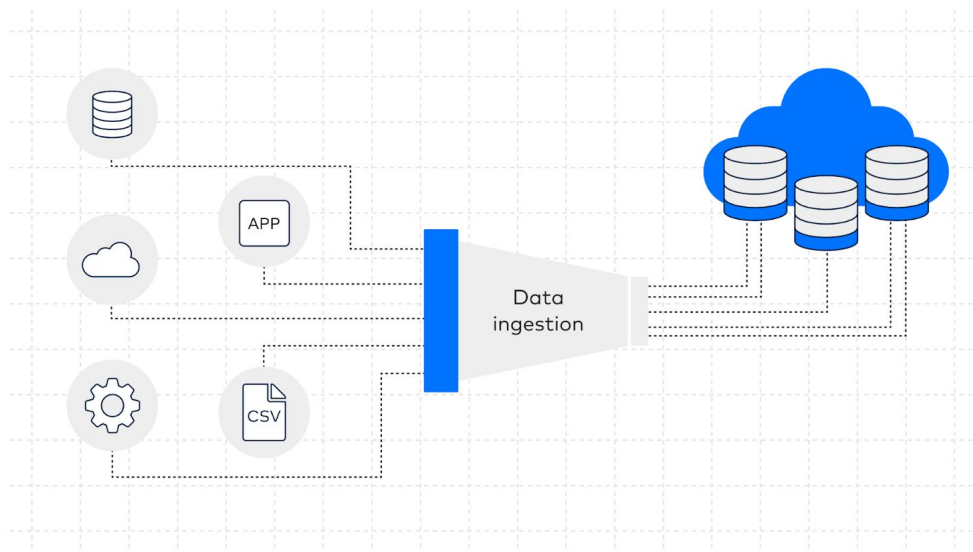
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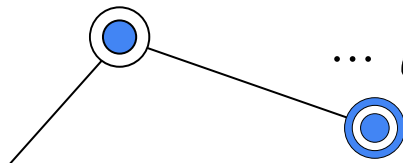
Data Ingestion:

- API's
- Public Datasets
- Vendors
- Clients

Data Storage:

- Local
- Cloud services

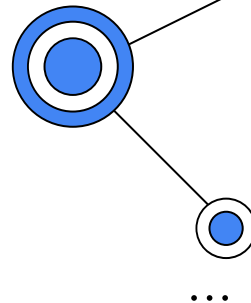




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Model Selection:

1. Set of Candidate Models
2. Training data, Validation data, testing data.
3. Evaluate Model Performance
4. Select the Best Model



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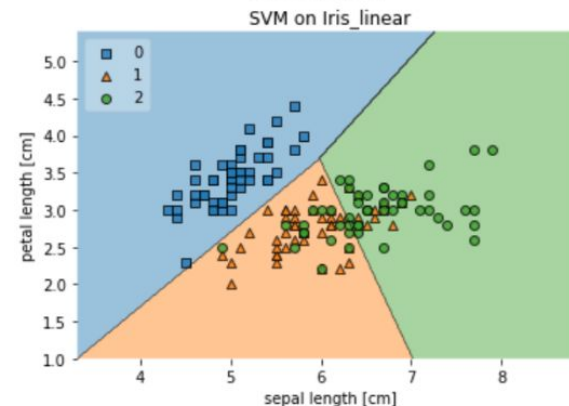
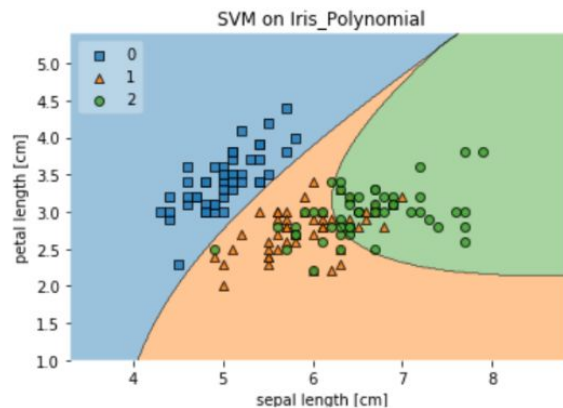
Hyperparameter Tuning:

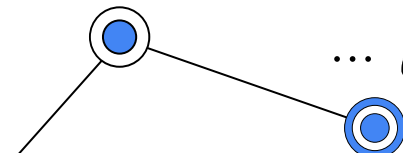
1. Grid Search, Random Search, Bayesian Optimization
2. Different combinations of HPs
3. Select the best combination

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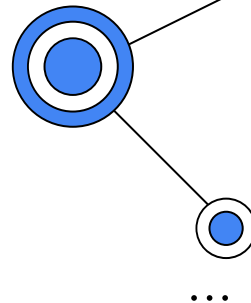
Evaluation: Polynomial kernel				
	precision	recall	f1-score	support
Iris-setosa	1.00	1.00	1.00	13
Iris-versicolor	1.00	0.88	0.93	8
Iris-virginica	0.90	1.00	0.95	9
accuracy			0.97	30
macro avg	0.97	0.96	0.96	30
weighted avg	0.97	0.97	0.97	30

Evaluation: linear kernel				
	precision	recall	f1-score	support
Iris-setosa	1.00	1.00	1.00	6
Iris-versicolor	1.00	1.00	1.00	9
Iris-virginica	1.00	1.00	1.00	15
accuracy			1.00	30
macro avg	1.00	1.00	1.00	30
weighted avg	1.00	1.00	1.00	30





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Model Deployment

- Model Serving - The process of addressing the ML model artifact in a production environment.
- Model Performance Monitoring - The process of observing the ML model performance based on live and previously unseen data, such as prediction or recommendation. In particular, we are interested in ML-specific signals, such as prediction deviation from previous model performance. These signals might be used as triggers for model re-training.
- Model Performance Logging - Every inference request results in the log-record.

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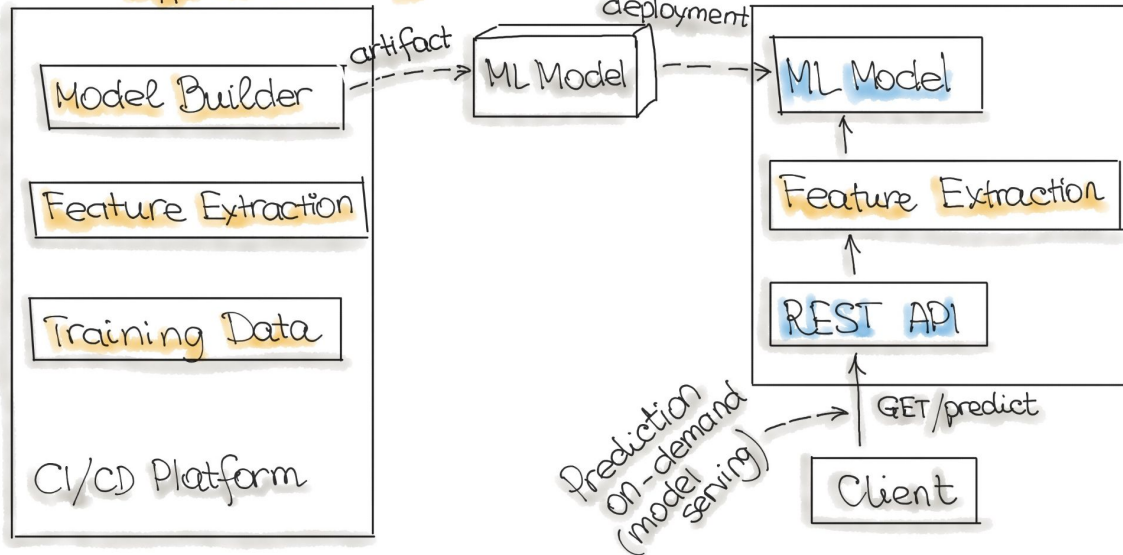
MODEL SERVING AS MICROSERVICE

TRAINING PHASE

PREDICTION PHASE

Development & Deployment
(offline learning)

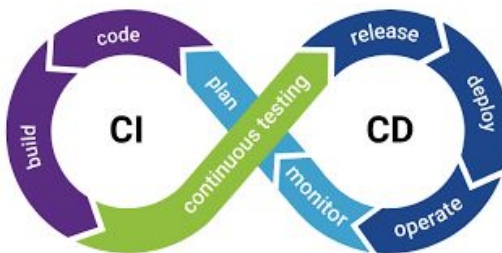
Production Environment



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CI/CD Pipelines

- Automating the deployment process with **CI/CD** (Continuous Integration/Continuous Deployment) pipelines is crucial for maintaining a machine learning (ML) system.
- This enables seamless updates to the code, model, and infrastructure with minimal manual intervention. By using **GitHub Actions** or **Jenkins**, you can automate running tests, deploying model updates, and even retraining the model periodically.



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CI/CD Pipelines

- Below are the key steps involved in setting up CI/CD pipelines for ML systems:
 - **Continuous Integration (CI)**
 - Run Unit Tests on New Code:
 - Every time a new commit or pull request is made to the repository, unit tests should be automatically run to ensure that the code changes don't break existing functionality.
 - You can use testing frameworks like pytest for Python code or unittest.
 - Linting and Code Style Checks: You can integrate tools like flake8 or black to ensure consistent code quality and style across your codebase.
 - Code Coverage: Track test coverage to ensure that all parts of the code are being tested.

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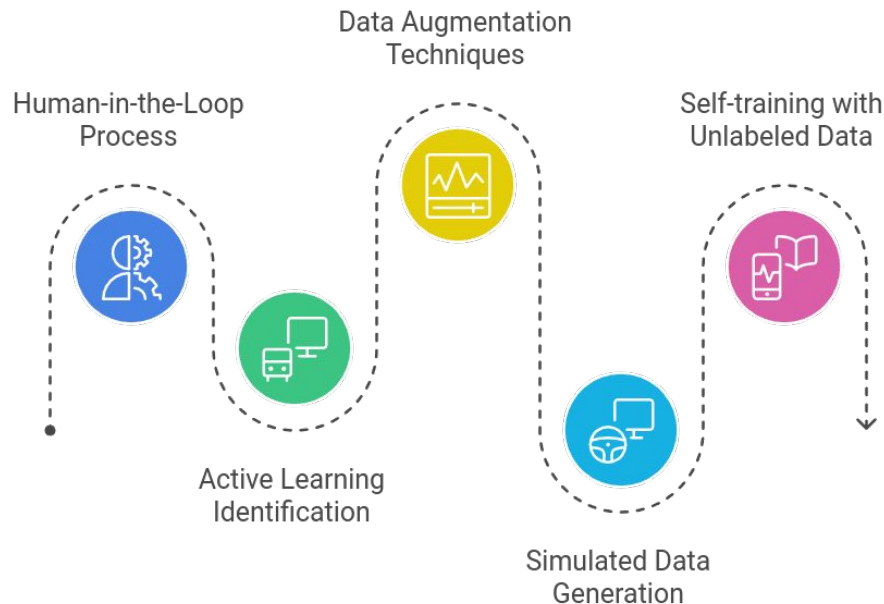
CI/CD Pipelines

- Below are the key steps involved in setting up CI/CD pipelines for ML systems:
 - **Continuous Deployment (CD)**
 - Once your tests pass, the next step is to deploy the code and model updates automatically.
 - Build and Deploy Model to the Hosting Platform: For model deployment, you can deploy to platforms like Streamlit Cloud, Heroku, AWS, or GCP. The deployment will involve pushing the updated model (saved as a pickle or other formats like TensorFlow SavedModel) to the cloud.
 - Model Versioning: It's important to manage model versions using a tool like MLflow. When deploying, always ensure the deployed version is logged so you can track which version of the model is in production.

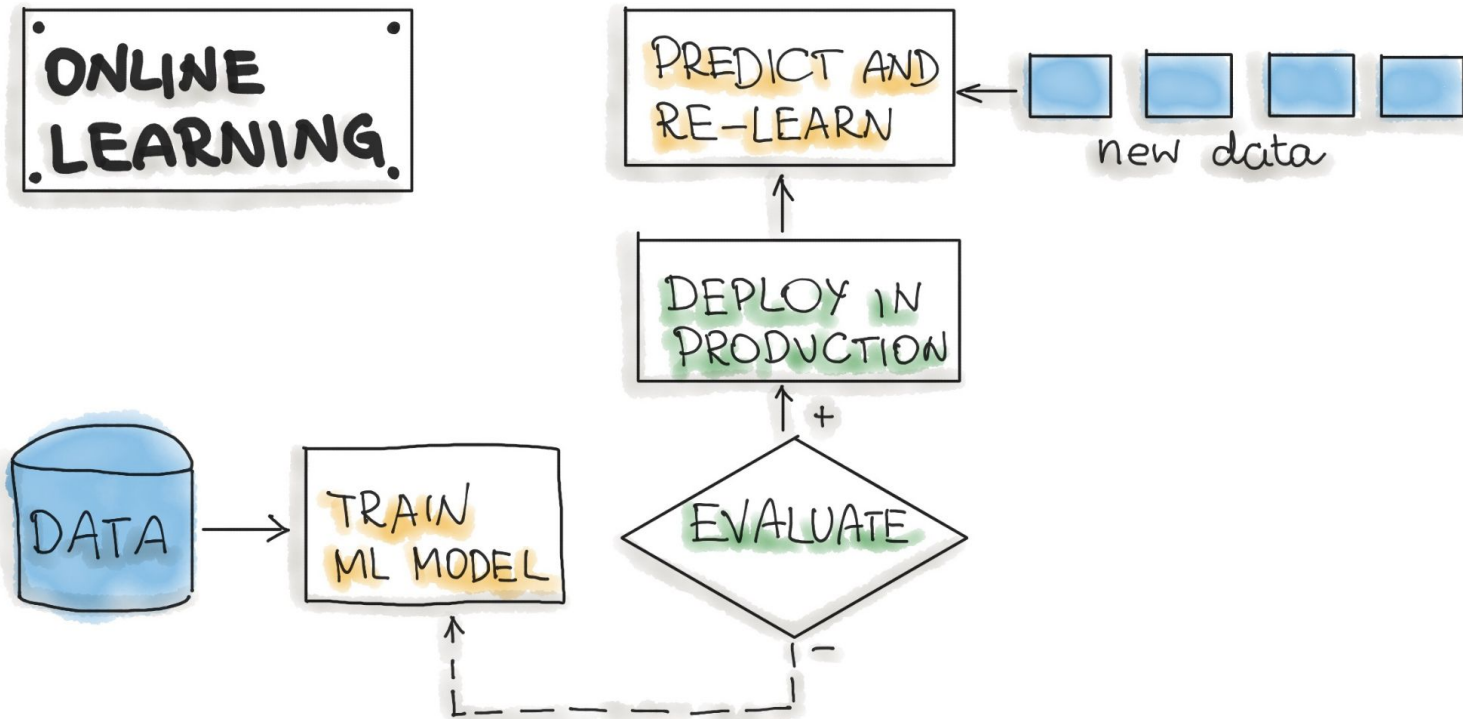
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Automatically retraining a **supervised machine learning** model typically involves **retraining the model** when new data becomes available.

However, since supervised learning **requires labeled data** for training, the **challenge** lies in ensuring that labels are either available or generated for the new data.



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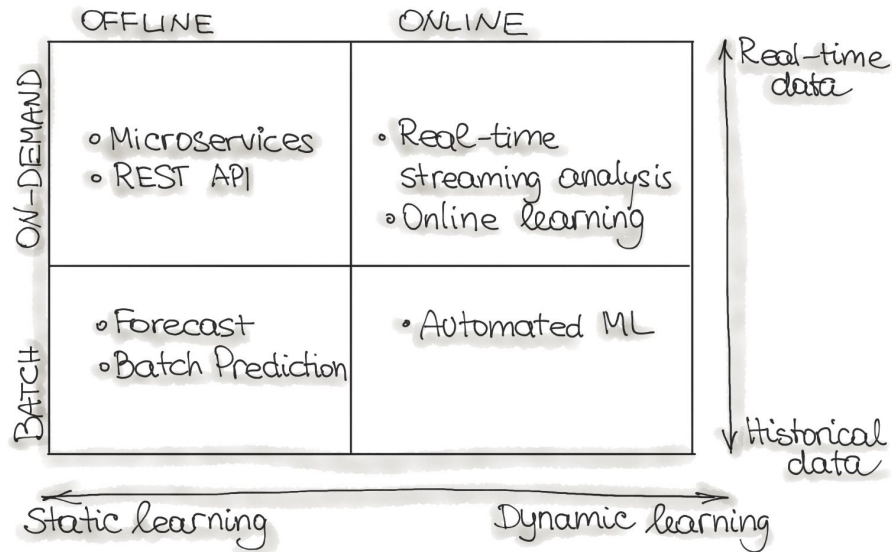


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MODEL SERVING PATTERNS

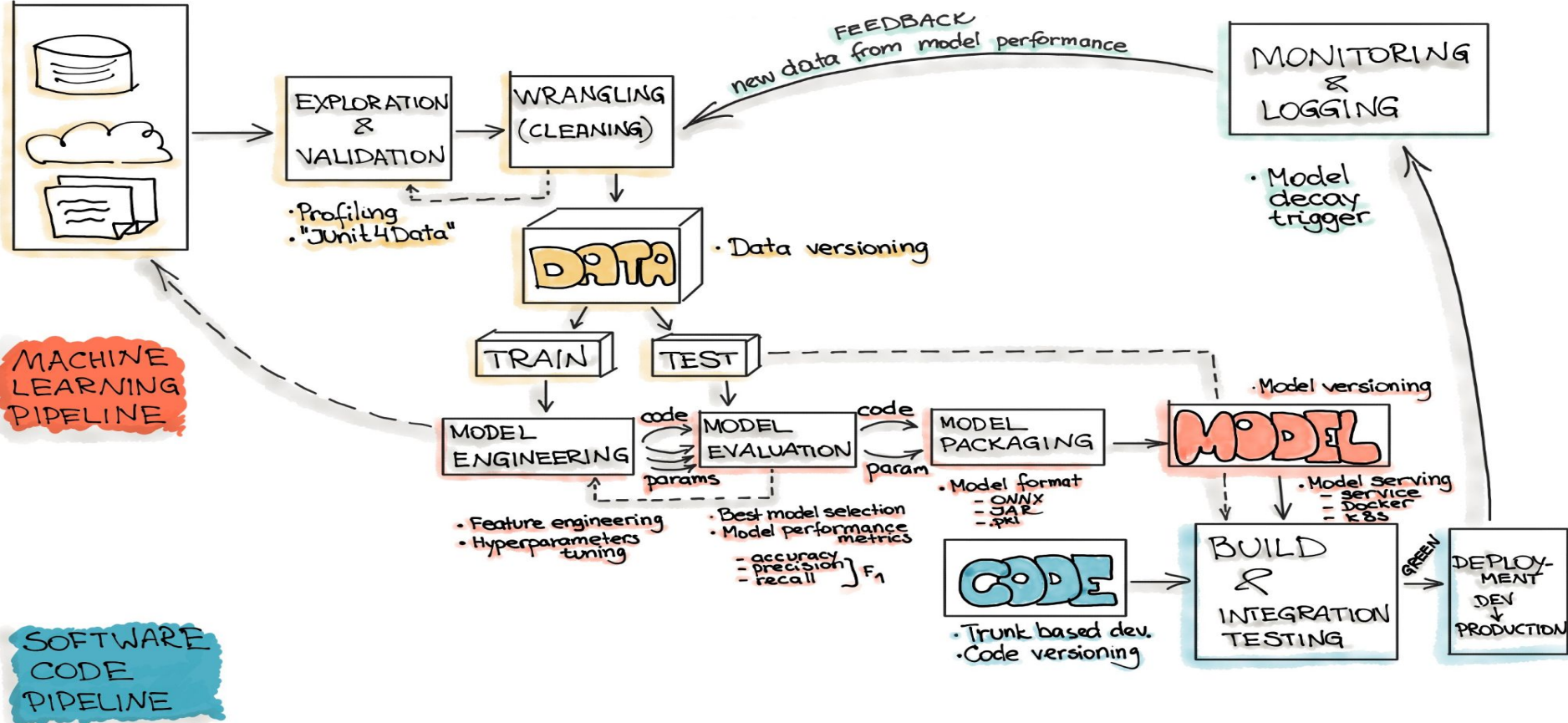
MODEL LEARNING

MODEL PREDICTION



DATA PIPELINE

MACHINE LEARNING ENGINEERING



L17 - MLOps Operations

Textbook

